In [4]:

```python
import numpy as np
import matplotlib.pyplot as plt

def main():
    X_mu = np.array([6, 4, 4, 5, 3, 4, 5, 0, 0, 0, 0]).reshape(-1, 1)
    N = len(X_mu)

    k = np.arange(N).reshape(-1, 1)
    mu = np.arange(N).reshape(1, -1)
    K = k @ mu

    W = np.exp(1j * (2 * np.pi / N) * K)

    x_k = (1 / N) * (W @ X_mu)

    print(f"Number of samples N = {N}\n")
    print("Matrix K:")
    print(K)
    print("\nMatrix W (rounded):")
    print(np.round(W, 2))

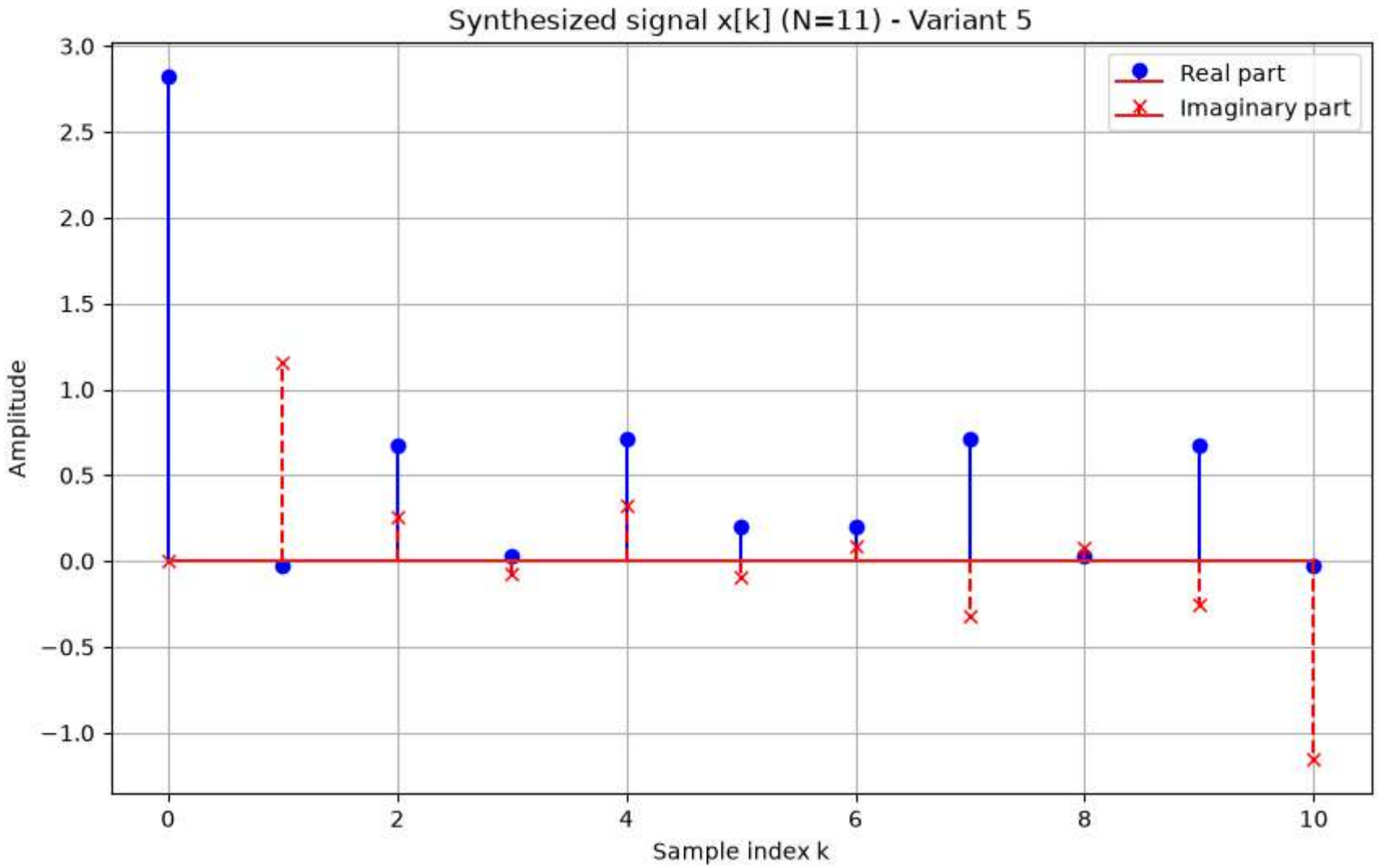
    plt.figure(figsize=(10, 6))
    plt.stem(np.arange(N), x_k.real.flatten(), linefmt='b-', markerfmt='bo', label='Real part')
    plt.stem(np.arange(N), x_k.imag.flatten(), linefmt='r--', markerfmt='rx', label='Imaginary part')
    plt.title(f'Synthesized signal x[k] (N={N}) - Variant 5')
    plt.xlabel('Sample index k')
    plt.ylabel('Amplitude')
    plt.legend()
    plt.grid(True)
    plt.show()

if __name__ == '__main__':
    main()
```

```
Number of samples N = 11

Matrix K:
[[  0   0   0   0   0   0   0   0   0   0   0]
 [  0   1   2   3   4   5   6   7   8   9  10]
 [  0   2   4   6   8  10  12  14  16  18  20]
 [  0   3   6   9  12  15  18  21  24  27  30]
 [  0   4   8  12  16  20  24  28  32  36  40]
 [  0   5  10  15  20  25  30  35  40  45  50]
 [  0   6  12  18  24  30  36  42  48  54  60]
 [  0   7  14  21  28  35  42  49  56  63  70]
 [  0   8  16  24  32  40  48  56  64  72  80]
 [  0   9  18  27  36  45  54  63  72  81  90]
 [  0  10  20  30  40  50  60  70  80  90 100]]

Matrix W (rounded):
[[ 1.  +0.j    1.  +0.j    1.  +0.j    1.  +0.j    1.  +0.j    1.  +0.j
   1.  +0.j    1.  +0.j    1.  +0.j    1.  +0.j    1.  +0.j  ]
 [ 1.  +0.j    0.84+0.54j  0.42+0.91j -0.14+0.99j -0.65+0.76j -0.96+0.28j
  -0.96-0.28j -0.65-0.76j -0.14-0.99j  0.42-0.91j  0.84-0.54j]
 [ 1.  +0.j    0.42+0.91j -0.65+0.76j -0.96-0.28j -0.14-0.99j  0.84-0.54j
   0.84+0.54j -0.14+0.99j -0.96+0.28j -0.65-0.76j  0.42-0.91j]
 [ 1.  +0.j   -0.14+0.99j -0.96-0.28j  0.42-0.91j  0.84+0.54j -0.65+0.76j
  -0.65-0.76j  0.84-0.54j  0.42+0.91j -0.96+0.28j -0.14-0.99j]
 [ 1.  +0.j   -0.65+0.76j -0.14-0.99j  0.84+0.54j -0.96+0.28j  0.42-0.91j
   0.42+0.91j -0.96-0.28j  0.84-0.54j -0.14+0.99j -0.65-0.76j]
 [ 1.  +0.j   -0.96+0.28j  0.84-0.54j -0.65+0.76j  0.42-0.91j -0.14+0.99j
  -0.14-0.99j  0.42+0.91j -0.65-0.76j  0.84+0.54j -0.96-0.28j]
 [ 1.  +0.j   -0.96-0.28j  0.84+0.54j -0.65-0.76j  0.42+0.91j -0.14-0.99j
  -0.14+0.99j  0.42-0.91j -0.65+0.76j  0.84-0.54j -0.96+0.28j]
 [ 1.  +0.j   -0.65-0.76j -0.14+0.99j  0.84-0.54j -0.96-0.28j  0.42+0.91j
   0.42-0.91j -0.96+0.28j  0.84+0.54j -0.14-0.99j -0.65+0.76j]
 [ 1.  +0.j   -0.14-0.99j -0.96+0.28j  0.42+0.91j  0.84-0.54j -0.65-0.76j
  -0.65+0.76j  0.84+0.54j  0.42-0.91j -0.96-0.28j -0.14+0.99j]
 [ 1.  +0.j    0.42-0.91j -0.65-0.76j -0.96+0.28j -0.14+0.99j  0.84+0.54j
   0.84-0.54j -0.14-0.99j -0.96-0.28j -0.65+0.76j  0.42+0.91j]
 [ 1.  +0.j    0.84-0.54j  0.42-0.91j -0.14-0.99j -0.65-0.76j -0.96-0.28j
  -0.96+0.28j -0.65+0.76j -0.14+0.99j  0.42+0.91j  0.84+0.54j]]
```


Synthesized signal x[k] (N=11) - Variant 5
Real part
Imaginary part
Amplitude
Sample index k
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
10